

GUANTANAMO BAY NAS, Cuba

WMO: 783670

Lat: 19.906N

Lon: 75.207W

Elev: 17

StdP: 101.12

Time Zone: -5.00 (CUB)

Period: 94-19

WBAN: 11706

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.5	19.3	12.8	9.3	22.0	14.0	10.0	22.5	11.2	26.3	10.3	27.3	4.2	340	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	34.0	26.2	33.3	26.1	32.9	26.0	27.9	31.7	27.5	31.4	27.1	31.1	7.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	22.4	30.3	26.2	21.7	30.0	25.9	21.2	29.8	89.3	31.9	87.1	31.4	85.2	31.1	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	9.4	8.5	DB	16.9	35.6	1.5	1.3	15.8	36.5	15.0	37.3	14.1	38.0	13.0	39.0	(4)
			WB	14.3	28.9	1.9	1.2	12.9	29.8	11.8	30.5	10.8	31.2	9.4	32.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	24.9	24.9	25.4	26.2	27.1	28.1	28.7	28.8	28.3	27.6	26.7	25.6	(6)
(7)		DBStd	1.79	1.27	1.19	1.24	1.07	1.11	1.10	0.95	0.93	0.95	0.97	1.26	1.31	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6156	461	417	476	486	532	542	581	582	549	546	500	484	(10)
(11)		CDD18.3	3114	202	184	218	236	273	292	323	323	299	288	250	226	(11)
(12)		CDH23.3	31958	1519	1487	1856	2259	2874	3469	4039	4009	3454	2956	2233	1803	(12)
(13)		CDH26.7	11964	411	387	504	709	1036	1426	1821	1789	1438	1092	779	573	(13)
(14)	Wind	WSAvg	4.1	4.3	4.4	4.4	4.4	4.0	3.8	4.2	4.2	3.8	3.8	4.0	4.1	(14)
(15)	Precipitation	PrecAvg	567	27	21	31	31	83	51	24	50	73	122	39	28	(15)
(16)		PrecMax	1130	263	112	139	136	477	204	104	145	457	504	138	174	(16)
(17)		PrecMin	236	0	0	0	0	1	0	0	0	1	5	2	0	(17)
(18)		PrecStd	228	46	26	34	35	99	47	21	36	101	117	35	36	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	31.1	31.5	31.9	32.9	33.8	34.6	34.7	34.4	33.9	33.2	32.0	(19)
(20)			MCWB	23.2	23.3	23.8	24.5	25.3	25.6	26.2	26.7	26.7	26.5	25.7	24.3	(20)
(21)		2%	DB	30.3	30.2	30.6	31.1	31.9	33.0	33.8	33.8	33.4	32.2	31.9	31.1	(21)
(22)			MCWB	23.3	23.1	23.7	24.4	25.0	25.6	25.8	26.6	26.6	25.8	25.4	24.1	(22)
(23)		5%	DB	29.7	29.6	29.8	30.5	31.2	32.1	33.1	33.1	32.2	31.8	31.1	30.3	(23)
(24)			MCWB	23.2	23.0	23.5	24.2	25.0	25.7	25.8	26.5	26.2	25.7	25.0	24.0	(24)
(25)		10%	DB	29.0	29.0	29.1	29.9	30.8	31.5	32.2	32.1	31.8	31.0	30.3	29.5	(25)
(26)			MCWB	23.0	23.0	23.3	23.9	24.9	25.6	25.7	26.0	26.1	25.6	24.7	23.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.6	25.1	27.0	26.9	27.7	28.1	28.4	28.4	28.4	28.0	27.2	26.4	(27)
(28)			MADB	28.9	28.9	28.1	29.8	30.4	31.5	32.3	32.6	32.2	31.5	30.6	29.6	(28)
(29)		2%	WB	24.7	24.5	25.1	26.1	26.9	27.3	27.5	27.7	27.6	27.2	26.5	25.5	(29)
(30)			MADB	28.5	28.6	29.0	29.5	30.0	31.1	31.8	32.1	31.5	30.7	30.3	29.4	(30)
(31)		5%	WB	24.1	24.0	24.5	25.3	26.2	26.7	27.0	27.2	27.1	26.6	25.9	24.8	(31)
(32)			MADB	28.2	28.3	28.5	29.0	29.6	30.8	31.5	31.7	31.1	30.3	30.0	29.1	(32)
(33)		10%	WB	23.6	23.5	24.0	24.7	25.6	26.2	26.5	26.7	26.6	26.1	25.3	24.3	(33)
(34)			MADB	27.8	27.9	28.1	28.6	29.3	30.5	31.2	31.3	30.7	30.0	29.3	28.5	(34)
(35)	Mean Daily Temperature Range	MDBR	8.6	8.6	8.1	7.7	7.3	7.1	7.6	7.6	7.6	7.4	8.0	8.4	(35)	
(36)		5% DB	MADB	9.2	9.1	8.6	8.1	7.7	8.3	8.3	8.1	8.3	8.6	9.0	(36)	
(37)			MCWBR	4.7	4.8	4.4	4.2	4.0	4.0	4.1	4.1	3.9	3.9	4.1	4.4	(37)
(38)		5% WB	MADB	8.5	8.5	7.9	7.6	7.1	7.3	7.7	7.8	7.7	7.7	8.1	8.5	(38)
(39)			MCWBR	5.0	5.2	4.8	4.2	3.9	4.0	4.3	4.2	4.0	3.9	4.2	4.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.376	0.383	0.392	0.422	0.450	0.493	0.502	0.468	0.428	0.403	0.390	0.386	(40)	
(41)		taud	2.529	2.498	2.469	2.397	2.348	2.214	2.174	2.305	2.437	2.513	2.514	2.513	(41)	
(42)		Ebn at Noon	902	915	916	886	851	808	801	834	869	883	880	878	(42)	
(43)		Edh at Noon	98	106	113	123	127	144	151	133	115	103	99	97	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.54	5.41	6.05	6.41	6.12	6.17	6.46	6.25	5.79	4.95	4.51	4.29	(44)	
(45)		RadStd	0.29	0.29	0.30	0.33	0.35	0.38	0.28	0.22	0.34	0.42	0.32	0.25	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	+1.66	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	+1.66	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page